

Explicable Artificial Intelligence for Affective Computing

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Artificial intelligence (AI) is increasingly tasked with recognizing and responding to human emotions, making affective computing one of its most consequential frontiers. As AI spreads into finance, policymaking, and mental health, the opacity of deep learning models raises urgent challenges for trust, accountability, and ethics. This special issue addresses explicability not just as algorithmic transparency, but as a paradigm integrating cognitive science, the humanities, and ethical foresight with technical innovation. Guided by the “Seven Pillars for the Future of AI”—multidisciplinarity, task decomposition, parallel analogy, symbol grounding, similarity measure, intention awareness, and trustworthiness—it envisions affective AI as a partner in meaning-making rather than a mere inference engine. The six featured articles span topics from depression detection and sentiment analysis to hate speech moderation and interpretable driving behaviors, advancing affective AI that is accurate, interpretable, and aligned with human dignity.

The field of artificial intelligence (AI) has seen remarkable progress, yet with this advancement comes an increasing demand for transparency and interpretability, particularly in the sensitive domain of affective computing. Affective computing, which deals with AI's ability to understand, process, and react to human emotions, raises unique ethical and societal challenges. As AI systems become more integrated into our daily lives, from investment¹ to policy-making² and from health care³ to mental health,⁴ the need to understand how they arrive at their conclusions is no longer a luxury but a necessity.

The “black box” nature of many deep learning models, while powerful, obscures their decision-making processes, creating a chasm of distrust for end-users, developers, and policymakers alike.⁵ This opacity is

especially problematic in emotional contexts where misinterpretation is highly sensitive.⁶ Explicability here goes beyond technical model interpretability: It calls for a conceptual shift toward designing AI as a partner in meaning-making, not just a tool for inference. Explaining affective AI means more than revealing its mechanics: It requires placing its decisions within the cultural, social, and ethical contexts of human emotion.⁷ In this sense, explicability is inherently multidimensional, encompassing the cognitive sciences, the humanities, and social values, as much as it encompasses engineering ingenuity.

The motivation of this special issue lies in addressing this multidimensional challenge. It seeks to convene research that not only advances the technical state of the art in affective computing and applications, but also interrogates the ways in which these systems can be made intelligible, trustworthy, and beneficial to humanity. Most contributions align with the “Seven Pillars for the Future of AI”⁸: multidisciplinarity, task decomposition, parallel analogy, symbol grounding, similarity measure, intention awareness, and trustworthiness. These pillars serve as guiding

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principles to foster systems that are not only functional but also epistemically and ethically robust.

By bringing together perspectives across disciplines, this special issue aims to expand the very meaning of explicability in AI from a narrow concern with algorithmic transparency to a holistic paradigm that acknowledges the complexity of human-machine interaction. Such a paradigm emphasizes that explicable AI in affective computing must account for human dignity, foster accountability, and promote interdisciplinary dialogue. In doing so, it highlights a central proposition that the future of AI is not determined by technical sophistication alone, but by its capacity to remain aligned with human values in the most emotionally charged and ethically sensitive domains.

This special issue received 18 submissions in total, of which six were accepted (33.3%). The accepted articles represent a diverse spectrum of approaches, methodologies, and perspectives, all united by the common goal of advancing explicability in affective AI. The contributions to this special issue collectively affirm that explicability in affective computing is not a singular achievement but a continuous project. It demands ongoing interdisciplinary efforts, the integration of ethical foresight into design, and a commitment to transparency that evolves alongside technical progress. As guest editors, our hope is that this special issue will not only inform but also inspire, serving as both a scholarly resource and a call to action for researchers, practitioners, and policymakers to advance explicable AI in ways that honor human dignity and foster collective well-being.

CONTENTS OF THE SPECIAL ISSUE

The first article, "Fine-Tuning Large Language Models with Behavioral Alignment for Depression Detection,"^{A1} presents a novel approach to depression detection on social media via the depression large language model (DLLM), a two-stage fine-tuning framework designed to improve accuracy, robustness, and interpretability in affective AI.

In the first stage, DLLM is fine-tuned with tailored prompts that integrate multimodal data, e.g., text, images, and behavioral metadata, to capture diverse signals of depressive states. The second stage introduces behavioral alignment, whereby the model learns to differentiate positive and negative user behaviors, enhancing generalization and sensitivity to subtle indicators of mental health. Extensive experiments on the WU3D dataset show DLLM outperforming strong baselines, achieving notable improvements. Beyond technical performance, the framework addresses the

broader challenge of explicability in affective computing by situating predictions within observable patterns of user behavior, bridging opaque deep learning and human-interpretable signals.

Next, "Considering Sentiment Causes in In-Context Learning for Aspect-Based Sentiment Analysis"^{A2} addresses the challenge of explicability in aspect-based sentiment analysis (ABSA) within the paradigm of in-context learning. While LLMs have shown strong performance in few-shot ABSA by leveraging demonstration examples, existing retrieval methods primarily emphasize semantic and syntactic similarity, often overlooking the deeper drivers of sentiment. To bridge this gap, the authors propose semantic-causal contextual demonstration retrieval (SCCDDr), a novel retriever that integrates semantic, syntactic, and aspect-sentiment relationships with explicit modeling of sentiment causes (see Figure 1). SCCDDr leverages contrastive learning to align candidate demonstrations with test samples across these dimensions, thereby enriching both prediction accuracy and interpretability. Extensive experiments on four benchmark datasets across two ABSA tasks—e.g., aspect-based sentiment classification and aspect extraction with sentiment classification—demonstrate that SCCDDr consistently outperforms traditional retrievers, such as BM25, BERT-PT, and Instructor. The approach not only enhances classification accuracy but also offers interpretable rationales by situating sentiment judgments in their underlying causes. By foregrounding the "why" behind sentiment expressions, this work advances explicability in affective computing and exemplifies how integrating linguistic, cognitive, and causal signals can enhance both AI trustworthy and performance.

The next work, "AI-Based Hate Speech Detection System Using Video URLs for Effective Content Moderation,"^{A3} tackles one of the most pressing ethical challenges in affective computing: the detection of gender-based hate speech in digital environments. The authors introduce an AI system, designed for identifying women-centric hate speech in low-resource languages, addressing both technical and societal needs. Their approach begins with the collection of more than 11,000 YouTube video URLs using targeted queries. From these, audio was extracted, transcribed into Urdu, and then transliterated into Roman Urdu, producing two distinct datasets. These corpora were carefully annotated with substantial interannotator agreement to ensure quality and reliability. The system combines classical machine learning and deep learning methods, with results showing that the BERT-based model achieved outstanding accuracy, reaching

0.96 on Roman Urdu data. The study highlights the broader social impact of explicable AI: By focusing on women-targeted hate speech in low-resource settings, it advances the creation of safer, more inclusive online spaces and demonstrates how AI can be aligned with human dignity and ethical responsibility.

The article "ARISE: Explainable Multimodal Aggressive Driving Detection via Driver State and Environment Perception"^{A4} addresses the urgent need for interpretable models in intelligent transportation systems. Aggressive driving is a major safety risk, yet existing detection methods largely focus on time-series vehicle dynamics and driver behavior while neglecting emotions and external conditions. To bridge this gap, the authors propose ARISE, a multisource aggregation framework that integrates physiological, behavioral, and emotional signals with vehicle sensor data and environmental descriptors. ARISE introduces three key innovations, namely: a motion similarity descriptor to capture temporal continuity in vehicle states, an aggression descriptor to quantify hostile driving environments, and a driving performance descriptor to evaluate workload and stability. The system leverages multimodal feature extraction and fusion to produce both accurate predictions and interpretable insights into driver–environment interactions. To support this work, the authors collected a novel multimodal dataset of aggressive driving under emergency conditions, incorporating physiological signals, vehicle dynamics, environmental factors, and self-reported emotions. Extensive experiments show that ARISE outperforms state-of-the-art transformer- and graph-based methods, achieving over 9% F1 improvement. By combining

technical rigor with explicability, ARISE advances affective AI for safer and more trustworthy intelligent driving systems. It also demonstrates how affective AI as a fundamental pillar of intelligent transportation lies in its ability to reconcile performance with human-centered transparency.

In "Explainable Sentiment Analysis With DeepSeek-R1: Performance, Efficiency, and Few-Shot Learning,"^{A5} the authors advance the growing discourse on explicable AI in affective computing by presenting the first comprehensive evaluation of DeepSeek-R1, an open source reasoning-based LLM, for sentiment and emotion analysis. The authors benchmark DeepSeek-R1, both its 671B full model and multiple distilled variants, against OpenAI's GPT-4o and GPT-4o-mini across Amazon Reviews (five-class sentiment), IMDB (binary sentiment), and GoEmotions (27-class emotion recognition). Results demonstrate that DeepSeek-R1 achieves superior accuracy and efficiency in few-shot learning, representing an eightfold efficiency gain over GPT-4o. Importantly, its reasoning-oriented architecture produces transparent, step-by-step traces, offering built-in explainability beyond post hoc methods. The study highlights that distillation quality depends more on architectural design than on sheer parameter size: for example, a 32B Qwen2.5-based model outperforms a 70B Llama-based counterpart. While DeepSeek-R1 incurs greater computational costs and lower throughput than proprietary models, its interpretability and nuanced handling of fine-grained sentiment distinctions make it a robust alternative. Collectively, the article demonstrates the tradeoffs between reasoning depth, efficiency, and

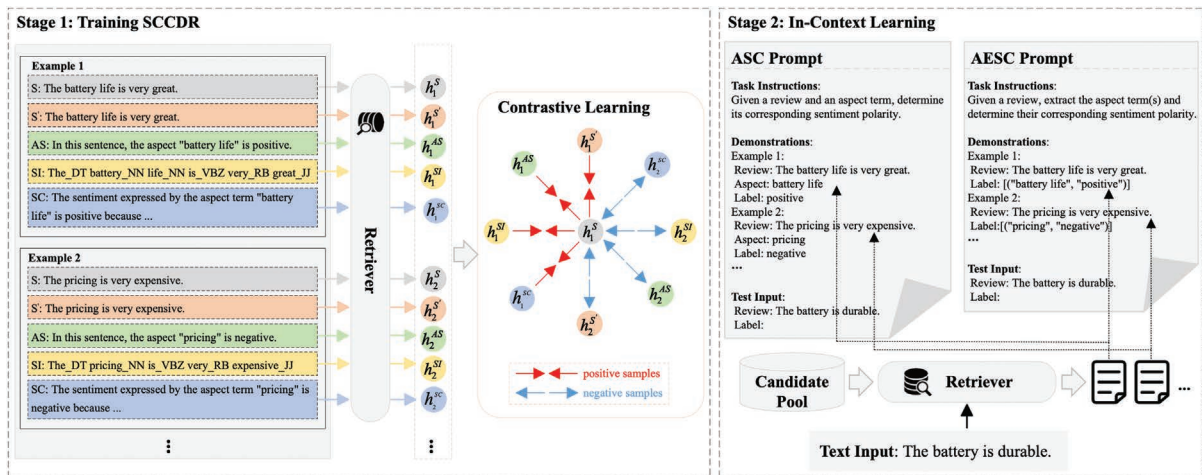


FIGURE 1. The framework proposed in the work "Considering Sentiment Causes in In-Context Learning for Aspect-Based Sentiment Analysis."^{A2}

accuracy, and positions open source reasoning LLMs as promising, trustworthy solutions for explainable sentiment analysis.

Finally, in "Leveraging ChatGPT-based Augmentation and Contrastive Learning for Chinese Massive Open Online Course Sentiment Analysis,"^{A6} the authors explore how contrastive learning and data augmentation can be jointly leveraged to advance sentiment analysis of Chinese massive open online course (MOOC) reviews. They propose integrating three augmentation strategies, namely similar word replacement, random word deletion, and ChatGPT-based augmentation (GPTaug), with a dual-loss framework combining cross-entropy and InfoNCE contrastive losses. Using a large dataset of more than 14,000

MOOC review comments, the study systematically investigates hyperparameter sensitivity, augmentation strategy effectiveness, and performance across sentiment categories. Results demonstrate that the integration of contrastive learning with data augmentation consistently improves classification accuracy, with GPTaug showing notable advantages in detecting neutral sentiments. Beyond technical contributions, the work highlights practical implications for monitoring learner satisfaction and guiding instructional improvements in online education. By demonstrating how explicable augmentation strategies can enhance robustness and interpretability in affective computing, the study offers both methodological insights and actionable pathways for educational applications.

APPENDIX: RELATED ARTICLES

- A1. X. Ning, H. Sun, D. Yu, C. Yang, R. Fang, and L. Fan, "Fine-tuning large language models with behavioral alignment for depression detection," *IEEE Intell. Syst.*, vol. 40, no. 6, pp. 10–18, Nov./Dec. 2025, doi: [10.1109/MIS.2025.3584888](https://doi.org/10.1109/MIS.2025.3584888).
- A2. M. Shi, R. Fan, T. He, and G. Chen, "Considering sentiment causes in in-context learning for aspect-based sentiment analysis," *IEEE Intell. Syst.*, vol. 40, no. 6, pp. 19–28, Nov./Dec. 2025, doi: [10.1109/MIS.2025.3584862](https://doi.org/10.1109/MIS.2025.3584862).
- A3. Z. A. Khan, Y. Xia, F. Khaliq, W. Jiang, and M. S. Anwar, "AI-based hate speech detection system using video URLs for effective content moderation," *IEEE Intell. Syst.*, vol. 40, no. 6, pp. 29–40, Nov./Dec. 2025, doi: [10.1109/MIS.2025.3594849](https://doi.org/10.1109/MIS.2025.3594849).
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